# Exercise Sheet 4

## Discriminative Models: Linear and Logistic Regression, Calibrated Scores

Probabilistic Machine Learning

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# 4.A Least Squares Is a Likelihood

# Exercise 4.A.1: Four Flats and a Villa

Model $y = w_0 + w_1 x + \varepsilon, \varepsilon \sim \mathcal{N}(0, \sigma^2)$ i.i.d.; $x$ in m², $y$ in k€.

| $x$ | 40 | 60 | 80 | 100 |
|---|---|---|---|---|
| $y$ | 150 | 200 | 280 | 330 |

1. Write down $\ell(w_0, w_1, \sigma^2)$ and show that for **every** $\sigma^2$ the maximum over $(w_0, w_1)$ is the least-squares fit.
2. Compute $\hat{w}_1 = S_{xy}/S_{xx}$ and $\hat{w}_0 = \bar{y} - \hat{w}_1 \bar{x}$, the residuals and the RSS.
3. Compute $\hat{\sigma}^2_{\mathrm{ML}} = \mathrm{RSS}/n$ and the unbiased $\mathrm{RSS}/(n-2)$. By which factor does the ML version underestimate $\sigma^2$ on average?
4. Give the plug-in 95% band $\hat{y} \pm 1.96\hat{\sigma}_{\mathrm{ML}}$ at 80 m² and at 600 m². Then use the half-width that counts the uncertainty about $\mathbf{w}$, $1.96\hat{\sigma}\sqrt{1 + \frac{1}{n} + \frac{(x-\bar{x})^2}{S_{xx}}}$. What do you conclude?

# Exercise 4.A.1: Four Flats and a Villa — Solution

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1. $\ell = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_i (y_i - w_0 - w_1 x_i)^2$. The first term has no $w$, and $\frac{1}{2\sigma^2} > 0$, so the arg max over $w$ is the arg min of the RSS, whatever $\sigma^2$ is.
2. $\bar{x} = 70, \bar{y} = 240, S_{xx} = 2000, S_{xy} = 6200$: $\hat{w}_1 = 3.1$ k€ per m², $\hat{w}_0 = 240 - 217 = 23$. Fitted $147, 209, 271, 333$; residuals $3, -9, 9, -3$; $\mathrm{RSS} = 180$.
3. $\hat{\sigma}^2_{\mathrm{ML}} = 45$ ($\hat{\sigma} \approx 6.7$); $\mathrm{RSS}/2 = 90$ ($\approx 9.5$). Since $E[\mathrm{RSS}] = (n-2)\sigma^2$, the ML version is on average $\frac{n-2}{n} = \frac{1}{2}$ of $\sigma^2$ — with 4 points and 2 weights, half the noise disappears into the fit.
4. Plug-in: $271 \pm 13$ at 80 m², $1883 \pm 13$ at 600 m². With the uncertainty about $\mathbf{w}$: factor $\sqrt{1.3} \approx 1.14$ at 80 m² ($\pm 15$), $\sqrt{141.7} \approx 11.9$ at 600 m² ($\pm 157$). The plug-in band is as narrow at the villa as at a flat; the honest one is twelve times wider — and even that one trusts the straight line out to 600 m².

# 4.B Classification: Only p(y | x)

# Exercise 4.B.1: Cross-Entropy and Its Gradient

$h_i = \sigma(\mathbf{w}^\top \mathbf{x}_i)$ with $\sigma(z) = 1/(1+e^{-z})$ and $\ell(\mathbf{w}) = \sum_i [y_i \ln h_i + (1-y_i)\ln(1-h_i)]$.

1. Show $\sigma'(z) = \sigma(z)(1-\sigma(z))$.
2. Show $\frac{\partial \ell}{\partial \mathbf{w}} = \sum_i (y_i - h_i)\mathbf{x}_i$.
3. Six transactions, $\mathbf{x} = (1, \text{new device})$: new device — fraud, fraud, legitimate; known device — fraud, legitimate, legitimate. Compute the gradient at $\mathbf{w} = 0$.
4. Set both components of the gradient to zero: which scores does the MLE give to new and to known devices? Give $\hat{w}_0$, $\hat{w}_1$ and read $\hat{w}_1$ as an odds ratio.

# Exercise 4.B.1: Cross-Entropy and Its Gradient — Solution

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1. $\sigma'(z) = \frac{e^{-z}}{(1+e^{-z})^2} = \frac{1}{1+e^{-z}} \cdot \frac{e^{-z}}{1+e^{-z}} = \sigma(z)(1-\sigma(z))$.
2. $\frac{\partial}{\partial \mathbf{w}}[y \ln h + (1-y)\ln(1-h)] = \left(\frac{y}{h} - \frac{1-y}{1-h}\right) h(1-h)\mathbf{x} = (y-h)\mathbf{x}$; sum over $i$. Error × feature, as in LMS.
3. At $\mathbf{w} = 0$ every $h_i = 0.5$. Intercept: $\sum_i (y_i - 0.5) = 3 - 3 = 0$. Slope: only new devices count, $0.5 + 0.5 - 0.5 = 0.5$. Gradient $(0, 0.5)$: raise the weight of "new device".
4. Slope component: $\sum_{\text{new}} (y_i - h_{\text{new}}) = 2 - 3h_{\text{new}} = 0$, so $h_{\text{new}} = 2/3$; then the intercept component gives $h_{\text{known}} = 1/3$ — each group's observed fraud rate. $\hat{w}_0 = \ln \frac{1/3}{2/3} = \ln \frac{1}{2} \approx -0.69$, $\hat{w}_1 = \ln 2 - \ln \frac{1}{2} = \ln 4 \approx 1.39$: a new device multiplies the odds of fraud by 4.

# Exercise 4.B.2: The Peak That Runs Away (Python)

```
import numpy as np
x = np.array([-2.0, -1.5, -1.1, -0.6, -0.3, 0.2, 0.6, 1.0, 1.4, 1.9])
y = np.array([0, 0, 0, 0, 0, 0, 1, 1, 1, 1])
sigma = lambda z: 1 / (1 + np.exp(-z))

def fit(tau2=None, steps=10_000, eta=0.05):
    w0 = w1 = 0.0
    for t in range(1, steps + 1):
        h = sigma(w0 + w1 * x)
        g0 = ___                        # gradient for the intercept
        g1 = ___                        # gradient for the slope
        if tau2 is not None:
            g1 -= ___                   # log-prior N(0, tau2) on the slope
        w0, w1 = w0 + eta * g0, w1 + eta * g1
        if t in (10, 100, 1_000, 10_000):
            h = sigma(w0 + w1 * x)
            print(t, round(w1, 2), np.sum(y * np.log(h) + (1 - y) * np.log(1 - h)))
```

1. Fill in the blanks and run `fit()`. What happens to the slope, the log-likelihood and the scores? Why will it never stop?
2. Run `fit(tau2=1)` and `fit(tau2=100, steps=100_000)`. Explain the difference with the MAP formula of the lecture.



# Exercise 4.B.2: The Peak That Runs Away — Solution

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```
g0 = np.sum(y - h)
g1 = np.sum((y - h) * x)
g1 -= w1 / tau2
```

1. Slope $1.37 \to 3.64 \to 8.32 \to 17.76$; log-likelihood $-2.68 \to -1.04 \to -0.36 \to -0.06$. The data are separable, so scaling up the weights always raises the likelihood towards 0: the scores approach exactly 0 and 1 and the peak does not exist — three heads in three flips.
2. $\tau^2 = 1$: slope $1.43$ from about 100 steps on, log-likelihood $-2.54$. The log-prior $-w_1^2/(2\tau^2)$ outgrows the vanishing gain in likelihood, so a finite MAP exists. $\tau^2 = 100$: a weaker prior, slope $7.83$ — finite, but steep. $\lambda = 1/\tau^2$ decides how far the peak is allowed to run.

# 4.C Generative vs Discriminative

# Exercise 4.C.1: The Same Line

Naive Bayes with yes/no flags, $\theta_{jy} = P(x_j = 1 \mid y)$, prior $\pi = P(y = 1)$. Lecture numbers: $\pi = 1/1000$; new device $0.6$ vs $0.1$; foreign IP $0.7$ vs $0.05$; night $0.5$ vs $0.2$ (fraud vs honest).

1. Show that $\log \frac{P(y=1|x)}{P(y=0|x)} = w_0 + \sum_j w_j x_j$ and give $w_j$ and $w_0$.
2. Compute the three weights and $w_0$, and check that a transaction with all three flags gets $P(\text{fraud} \mid x) \approx 17.4\%$.
3. The foreign-IP flag is **missing**. Compare what naive Bayes should do with plugging $x_{\text{IP}} = 0$ into the linear form.
4. Why can logistic regression, which has the same linear form, not do what naive Bayes does in 3?

# Exercise 4.C.1: The Same Line — Solution

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**1.** $\log p(x_j \mid y) = x_j \log \theta_{jy} + (1 - x_j) \log(1 - \theta_{jy})$. Take the difference between the classes and sum over $j$:

$$w_j = \log \frac{\theta_{j1}}{\theta_{j0}} - \log \frac{1 - \theta_{j1}}{1 - \theta_{j0}}, \qquad w_0 = \log \frac{\pi}{1 - \pi} + \sum_j \log \frac{1 - \theta_{j1}}{1 - \theta_{j0}}$$

**2.** $w_{\text{dev}} = \log 6 - \log \frac{0.4}{0.9} = 1.792 + 0.811 = 2.603$; $w_{\text{IP}} = \log 14 - \log \frac{0.3}{0.95} = 2.639 + 1.153 = 3.792$; $w_{\text{night}} = \log 2.5 - \log \frac{0.5}{0.8} = 0.916 + 0.470 = 1.386$. $w_0 = \log \frac{1}{999} - 0.811 - 1.153 - 0.470 = -9.340$. All flags: $-9.340 + 7.781 = -1.560$, $\sigma(-1.560) = 0.174$ ✓ — the same $210/1{,}209$ as in the lecture.

**3.** Missing: drop the IP factor entirely (it sums to 1 over both values), odds $6 \cdot 2.5/999 = 15/999$, $P \approx 1.5\%$. Plugging in $x_{\text{IP}} = 0$ instead claims "domestic IP observed": it keeps the factor $0.3/0.95$ (hidden in $w_0$), odds $4.74/999$, $P \approx 0.47\%$ — a third of the honest answer.

**4.** Logistic regression models only $p(y \mid x)$; it has no $p(x_{\text{IP}} \mid y)$ to sum over. Its $w_0$ is not split into per-attribute pieces, so there is nothing to remove — it must be given some value of $x_{\text{IP}}$ or be refitted without the feature.

# Exercise 4.C.2: Counting Twice, in Code

Complete the three marked lines (numpy only), then run the script.

```
import numpy as np
rng = np.random.default_rng(42)
n = 5000
y = (rng.random(n) < 0.3).astype(int)                        # 1 = fraud
flag = lambda pf, ph: (rng.random(n) < np.where(y == 1, pf, ph)).astype(int)
device = flag(0.6, 0.1)
copy = np.where(rng.random(n) < 0.95, device, 1 - device)   # a near-copy
night = flag(0.5, 0.2)
X = np.column_stack([device, copy, night])

def fit_nb(X, y, alpha=1.0):
    prior = ...                     # (a) P(y = 1)
    th1 = ...                       # (b) Laplace-smoothed P(x_j = 1 | y = 1)
    th0 = ...                       #     and P(x_j = 1 | y = 0)
    w = np.log(th1 / th0) - np.log((1 - th1) / (1 - th0))
    w0 = np.log(prior / (1 - prior)) + np.log((1 - th1) / (1 - th0)).sum()
    return w0, w

def fit_lr(X, y, steps=5000, lr=0.5):   # gradient ascent, as in the lecture
    Xb = np.column_stack([np.ones(len(X)), X]); w = np.zeros(Xb.shape[1])
    for _ in range(steps):
```

1. Print $w_0$, $w$ and $P(\text{fraud} \mid x = (1, 1, 1))$ for both models.
2. Compare the weights of `device` and `copy`. Which model counts the device twice, and which shares the credit?
3. Which probability is closer to the fraction of frauds among the training rows with $x = (1, 1, 1)$?

# Exercise 4.C.2: Counting Twice, in Code — Solution

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```
prior = y.mean()
th1 = (X[y == 1].sum(axis=0) + alpha) / ((y == 1).sum() + 2 * alpha)
th0 = (X[y == 0].sum(axis=0) + alpha) / ((y == 0).sum() + 2 * alpha)
```

**1.** (numpy 2.x, seed 42)

| model | $w_0$ | $w_{\text{device}}$ | $w_{\text{copy}}$ | $w_{\text{night}}$ | $P(\text{fraud} \mid 1,1,1)$ |
|---|---|---|---|---|---|
| naive Bayes | −2.80 | 2.60 | 2.17 | 1.35 | 0.965 |
| logistic regression | −2.10 | 2.75 | −0.13 | 1.40 | 0.873 |

**2.** Naive Bayes gives the copy almost the device's weight: the same evidence enters twice. Logistic regression fits all weights jointly, sees that the copy adds nothing once the device is known, and drives its weight to about 0.

**3.** Among the 486 training rows with $x = (1, 1, 1)$, 88.5% are fraud; the true posterior of the generating model is $0.865$. Logistic regression (0.873) is close; naive Bayes (0.965) is **over-confident** — the double counting of Lecture 03, measured.

# 4.D When Is a Score a Probability?

# Exercise 4.D.1: The Rebalanced Fraud Model

A model is trained on a 50/50 sample; on real traffic fraud is **1 in 1,000**. Undersampling kept every fraud and a random share of legitimate cases.

1. Use Bayes' theorem in odds form to show: real odds = model odds × $\frac{1/999}{1}$. Which assumption about $p(\mathbf{x} \mid y)$ do you need?
2. Convert the scores $0.5$, $0.9$ and $0.99$ into real fraud probabilities.
3. Blocking a legitimate card costs **€10**, missing a fraud **€500**. Above which real probability should a card be blocked, and which **raw score** is that?

# Exercise 4.D.1: The Rebalanced Fraud Model — Solution

# Exercise 4.D.1: The Rebalanced Fraud Model — Solution

1. Posterior odds = $\frac{p(\mathbf{x}|\text{fraud})}{p(\mathbf{x}|\text{legit})}$ × prior odds. Sampling that depends only on $y$ leaves $p(\mathbf{x} \mid y)$ unchanged, so the likelihood ratio is the same in both worlds; only the prior odds change, from $1$ (training) to $1/999$ (real).
2. $0.5$: odds $1/999$, $p = 0.10\%$. $0.9$: odds $9/999$, $p = 9/1008 \approx 0.89\%$. $0.99$: odds $99/999$, $p = 99/1098 \approx 9.0\%$.
3. Block if $500p > 10(1-p)$, i.e. $p > \frac{10}{510} \approx 1.96\%$ (real odds $> 0.02$). Raw odds $> 0.02 \cdot 999 = 19.98$, raw score $> 19.98/20.98 \approx 0.95$. A threshold of 0.5 on the raw score would block cards whose real risk is 0.1%.

# Exercise 4.D.2: Weight Decay Is a Prior

1. With $y_i = \mathbf{w}^\top \mathbf{x}_i + \varepsilon_i, \varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ and a prior $\mathbf{w} \sim \mathcal{N}(0, \tau^2 \mathbf{I})$, show that the MAP minimises $\mathrm{RSS}(\mathbf{w}) + \lambda \|\mathbf{w}\|^2$ with $\lambda = \sigma^2/\tau^2$.
2. One weight, no intercept, $y = wx + \varepsilon$. Show $\hat{w}_{\mathrm{MAP}} = \dfrac{\sum_i x_i y_i}{\sum_i x_i^2 + \lambda}$.
3. Data $(x, y) = (1, 2), (3, 7)$, $\sigma^2 = 1, \tau^2 = 0.1$. Compute the MLE and the MAP. What happens as $\tau^2 \to \infty$ and as $\tau^2 \to 0$?

# Exercise 4.D.2: Weight Decay Is a Prior — Solution

# Exercise 4.D.2: Weight Decay Is a Prior — Solution

1. $\log p(\mathbf{w} \mid D) = -\frac{\mathrm{RSS}(\mathbf{w})}{2\sigma^2} - \frac{\|\mathbf{w}\|^2}{2\tau^2} + \mathrm{const}$. Multiplying by $-2\sigma^2$ turns the maximum into the minimum of $\mathrm{RSS} + \frac{\sigma^2}{\tau^2}\|\mathbf{w}\|^2$.
2. The derivative of $\sum_i (y_i - w x_i)^2 + \lambda w^2$ is $-2\sum_i x_i y_i + 2w\sum_i x_i^2 + 2\lambda w$; zero at the formula.
3. $\sum x_i y_i = 23, \sum x_i^2 = 10, \lambda = 10$. MLE $23/10 = 2.3$; MAP $23/20 = 1.15$. $\tau^2 \to \infty$: $\lambda \to 0$, the MLE (a flat prior). $\tau^2 \to 0$: $\lambda \to \infty$, $\hat{w} \to 0$ — the prior overrules two data points, as a strong Beta prior did in Lecture 02.